

RESEARCH INTERESTS

- (Deep) (Model-Based) Reinforcement Learning Representation Learning

EDUCATION

- **Université de Montréal / Mila** Montréal, Canada
Master of Science in Computer Science Jan. 2021 – Present
 - **Supervisor:** Prof. Sarath Chandar
 - **Relevant Courses:** Representation Learning (A+), Continual learning (A+), Reinforcement Learning (A+), Self-supervised Representation Learning (A)
- **Sharif University of Technology** Tehran, Iran
Bachelor of Science in Computer Science Sep. 2015 – Sep. 2020

RESEARCH EXPERIENCE

- **Mila** Montréal, Canada
Thesis Based M.Sc. Student Jan. 2021 – Present
 - **Supervisor:** Prof. Sarath Chandar
 - **Focus:** Deep Model-Based Reinforcement Learning
- **Chair for CAMPAR** Munich, Germany
Undergraduate Research Assistant Summer 2019
 - **Supervisor:** Prof. Dr. N. Navab
 - **Worked on:** 3D camera re-localization in changing indoor environments

PUBLICATIONS

- *Replay Buffer with Local Forgetting for Adaptive Deep Model-Based Reinforcement Learning*
- **Rahimi-Kalahroudi, A.**, Rajendran, J., Momennejad, I., Van Seijen, H., and Chandar, S.
 Currently Under Review
- *Towards Evaluating Adaptivity of Model-Based Reinforcement Learning Methods*
- Wan, Y.* , **Rahimi-Kalahroudi, A.***, Rajendran, J., Momennejad, I., Chandar, S., and Van Seijen, H.
 International Conference on Machine Learning (**ICML**), 2022 (arXiv/code)

* denotes equal contribution.

WORKING EXPERIENCE

- **iCAD Dental** Ottawa, Canada
ML Research Developer (part-time) Jan. 2022 – May. 2022
- **CafeBazaar Inc.** Tehran, Iran
Data Scientist Sep. 2017 – Jan. 2021
 - **Teams:** Spam Detection on Applications Reviews, Application Recommender System, Search-Ads, Search Engine, User Profiling

ACADEMIC AND INDUSTRIAL PROJECTS

- **RLHive:** A research framework for reinforcement learning that contains the most well-known single-agent and multi-agents RL algorithms in PyTorch. (code , docs)
- **Camera Pose Generator for 3RScan Data:** As a part of working for 3D camera relocalization research project, this task is designed to generate different valid poses with their rendered scene. Rendering is done with OpenGL in NanoGUI environment.
- **Leader of Data Science in Search-Ads Team:** Led data science, mechanism design, and optimization problems in Search-ads team of CafeBazaar.

- **Image Style Transfer:** PyTorch implementation of image style transfer using CNNs. (code)
- **Online Scalable App-Recommender System:** Implemented model trainer, and also hyper-parameter tuning on a single GPU for online scalable android applications Rec-Sys of CafeBazaar using the implicit-als algorithm as being a member of the team. Final output of the team could handle over 500 requests per second.

HONORS AND AWARDS

- Bourse D'excellence DIRO 2022
- Université de Montréal Montréal, Canada
- M.Sc. Tuition Exemption Scholarships (Bourse C) 2021
- Université de Montréal Montréal, Canada
- International Undergraduate Excellence Award 2019
- Chair for CAMPAR of Technical University of Munich Munich, Germany
- Silver Medal 2014
- 24th Iranian National Olympiad in Informatics(INOI) Tehran, Iran

TEACHING EXPERIENCE

- **Sharif University of Technology** Tehran, Iran
Teacher Assistant
 - **Courses:** Advanced Programming, Data Structures
- **Iranian National Olympiad in Informatics** Tehran, Iran
Part of Scientific Committee for the Summer and Final Round Exams 2016 – 2018
- **High Schools** Tehran, Iran
Instructing Informatics Olympiad Materials 2015 – 2018

PROGRAMMING SKILLS

- **Languages:** Skilled in Python, Familiar with C++, Java, Scala, and L^AT_EX
- **ML and Data Science Tools:** Skilled in PyTorch, NumPy, Familiar with Tensorflow
- **Technologies:** Apache Spark (and PySpark), Apache Hadoop, Docker, Singularity